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Experimental report

Experiment: Positioning of drug moiety for TOUCANs

Date: 08.02.2022

By: Baris Ata Borsa

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Experimental Conditions

Recording Report (96-Well Plate Reader)

Experiment Overview:

- **Study Title:** Positioning of drug moiety for TOUCANs
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- **Experiment Date and time:** 08.02.2022
- **Laboratory:** Microbiology Lab at 2E:686
- **Type of Lab:** BSL-2
- **Lab Technician:** Baris Ata Borsa
- **Methodology:** Broth microdilution (BMD)
- **Measurement Type:** Naked eye and OD600 measurements
- **Protocol:** SOP-TL-AST

Other information: The aim of the experiment is to show the effective position of drug moiety for TOUCAN approach.

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Sample Type:

- **Sample 1:** *Staphylococcus aureus* ATCC29213

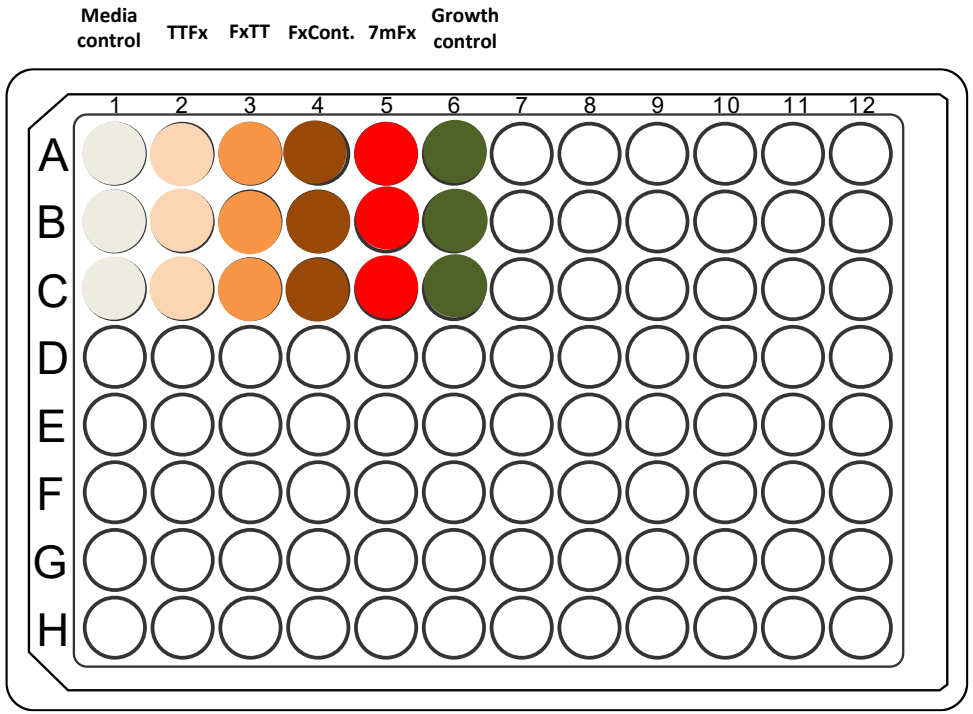
Drug and concentrations used:

- 7mFx (TOUCAN-MRSA): mU-mU-mU-mU-**T-T-Fx**
- FxTT: mU-mU-mU-mU-**Fx-T-T**-mU-mU-mU-mU
- FxControl: mU-mU-mU-mU-**Fx**-mU-mU-mU-mU
- TTFx: mU-mU-mU-mU-**T-T-Fx**-mU-mU-mU-mU

➢ Final concentration was 40 µg/ml for all tested drugs.

Experiment design

Plate Layout (96-Well Format):



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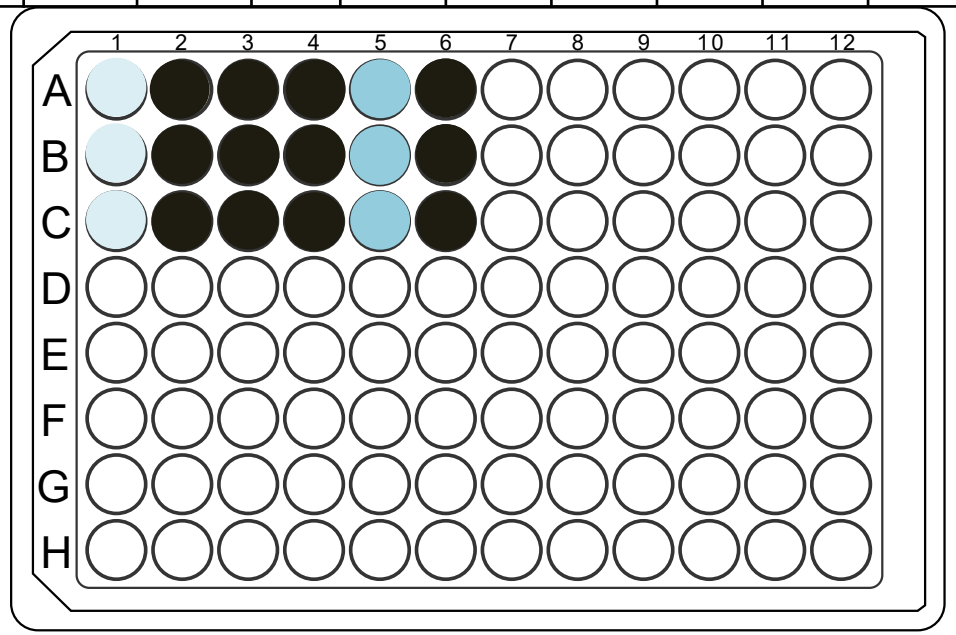
Data acquisition

Raw Data: OD600 at 18h

Media control	TTFx	FxTT	FxCont.	7mFx	Growth control										
0,086	1,204	1,569	1,584	0,257*	1,574										
0,087	1,263	1,511	1,508	0,352*	1,550										
0,085	1,299	1,580	1,617	0,261*	1,628										

Raw Data: Naked eye inspection



**See the notes and color codes below..*

Growth

No Growth

Notes:

Limited growth was observed (86,3% inhibition by OD600). This type of growth ("tiny buttons") is disregarded for Gram-positive bacteria with bacteriostatic antibiotics based on EUCAST recommendations (see below).

https://www.eucast.org/fileadmin/src/media/PDFs/EUCAST_files/MIC_testing/Reading_guide_BMD_v_5.0_2024.pdf

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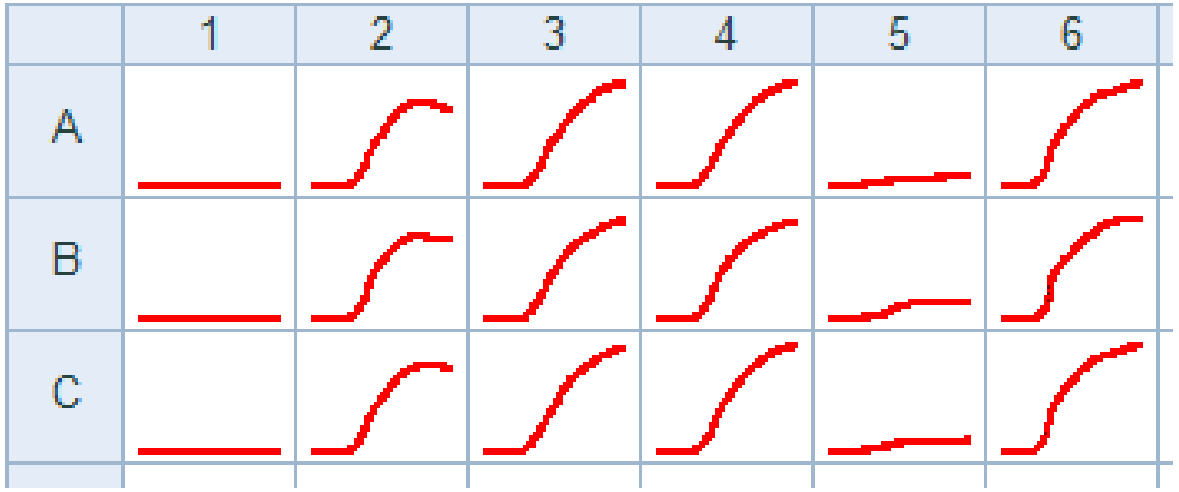
Data acquisition

Raw Data:
Growth curves from the device.



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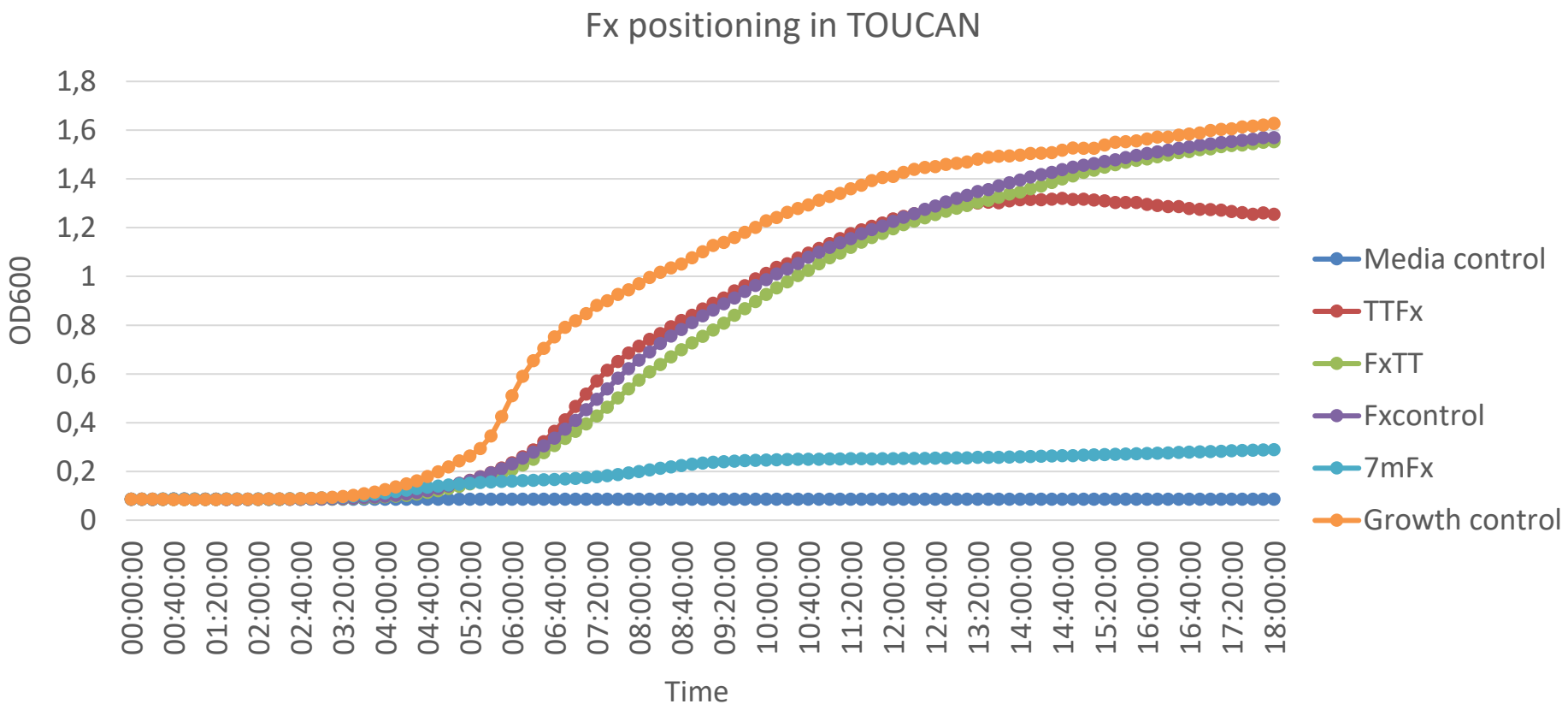
Data visualization

- Graphic for **average OD600 values over 18h** are presented below.



Date: 09.02.2022
Signature

Researcher:
Baris Ata Borsa

Date:
Signature

Supervisor:

Date:
Signature

Principal Investig.:
Frank J Hernandez

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